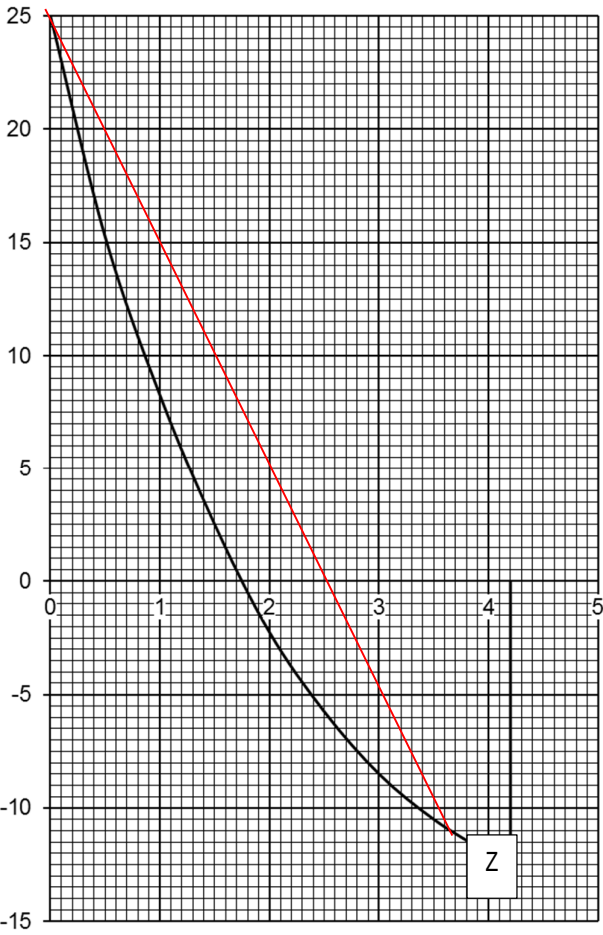
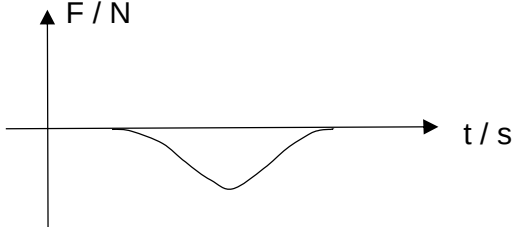


7	(a)	(i)	vertical height from ball launcher $= (1 \times 8) + \frac{1}{2} (0.8)(25-8) + 18 (0.1 \times 0.5) = 15.7 \text{ m}$ Vertical height from the level ground = $15.7 + 0.5 = 16.2 \text{ m}$	M1 A1
		(ii)	Acceleration at $t = 1.0 \text{ s}$ = gradient at $t = 1.0 \text{ s}$ $= -12.1 \text{ m s}^{-2}$ Using F2L, $mg + F_{\text{air}} = ma$ $F_{\text{air}} = ma - mg$ $= 0.320 (12.1 - 9.81)$ $= 0.733 \text{ N}$	M1 A1
		(iii) 1.	 Gradient = 9.8 m s^{-2} Negative gradient	A1 A1
		2.	At $t = 1.0 \text{ s}$, vertical velocity = 15.25 m s^{-1} since horizontal velocity = 12.0 m s^{-1}	M1

			Resultant velocity = $\sqrt{15.25^2 + 12^2} = 19.4 \text{ m s}^{-1}$	A1
	(b)	(i)	Consider a collision that occurs when X collides with Y in a straight line. By Newton's second law the change in momentum for X, $\Delta p_x = F_{yx} \Delta t$, where F_{yx} is the force Y exerts on X and Δt is the duration the force is exerted while the change in momentum for Y, $\Delta p_y = F_{xy} \Delta t$, where F_{xy} is the force X exerts on Y. By Newton's third law, $F_{yx} = -F_{xy}$ since they are an action-reaction pair. Hence, $\Delta p_x = -\Delta p_y$. This implies $p_{xF} - p_{xI} = -(p_{yF} - p_{yI})$, where p_{xF} is the final momentum of X, p_{xI} is the initial momentum of X, p_{yF} is the final momentum of Y and p_{yI} is the initial momentum of Y. Rearranging, $p_{xI} + p_{yI} = p_{xF} + p_{yF}$. This implies the total initial momentum is the same as the total final momentum if no external force acts on this system.	B1 B1 B1
		(ii)	1. $m_x u_x + m_y u_y = m_x v_x + m_y v_y$ $50(4.5) + m_y(-2.8) = 50(-1.8) + m_y(1.4)$ $m_y(1.4+2.8) = 50(4.5+1.8)$ $m_y = 75 \text{ g} = 0.075 \text{ kg}$	M1 C1 A1
			2. total initial kinetic energy/KE not equal to the total final kinetic energy/KE or relative speed of approach is not equal to relative speed of separation, showing calculation – e.g. relative speed of approach = $4.5 - (-2.8) = 7.3$ while relative speed of separation = $1.4 - (-1.8) = 3.2$ so not elastic or is inelastic	M1 A1
		(iii)	1.  According to Newton's 3rd Law, the forces exerted on the bodies by each other are equal in magnitude and act in opposite directions.	B1 B1
			2. Change in momentum of Y or impulse on Y	A1

2	(c)	(i)	The product of the magnetic flux density normal to the surface and the area of	
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